

Fundamentals of Electrical and Electronics Engineering (FEEE)

SUB CODE: ECE1002

T-P-C: 3-2-4

Lecture-8

Faculty

Dr. Rajasekhar N V

Associate Professor G-1

SENSE, VIT-AP University

Mail: rajasekhar.venkata@vitap.ac.in

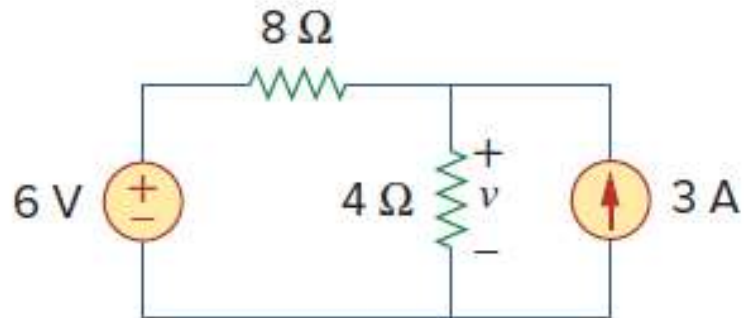
Cabin: Faculty Area 424-C:AB-1

Network Theorems

1. Superposition Theorem
2. Thevenin Theorem
3. Maximum Power Transfer Theorem

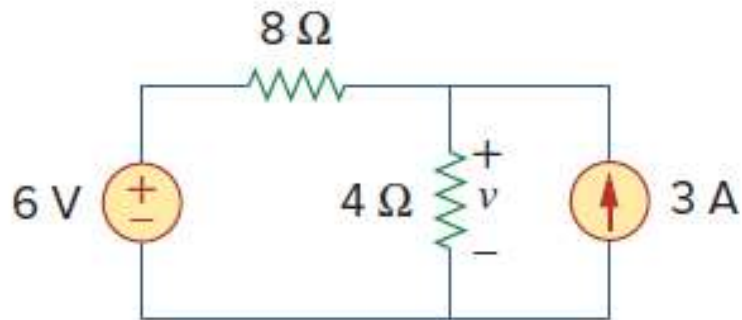
1. Superposition Theorem

- The superposition principle states that the voltage across (or current through) an element in a linear circuit is the algebraic sum of the voltages across (or currents through) that element due to each independent source acting alone.

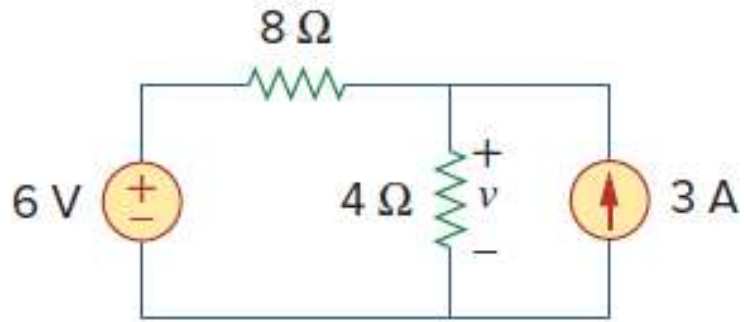


- Procedure:

1. We consider one independent source at a time while all other independent sources are turned off. This implies that we replace every voltage source by 0 V (or a short circuit), and every current source by 0 A (or an open circuit). This way we obtain a simpler and more manageable circuit.
2. Dependent source is left as it is because they are controlled by circuit variables.

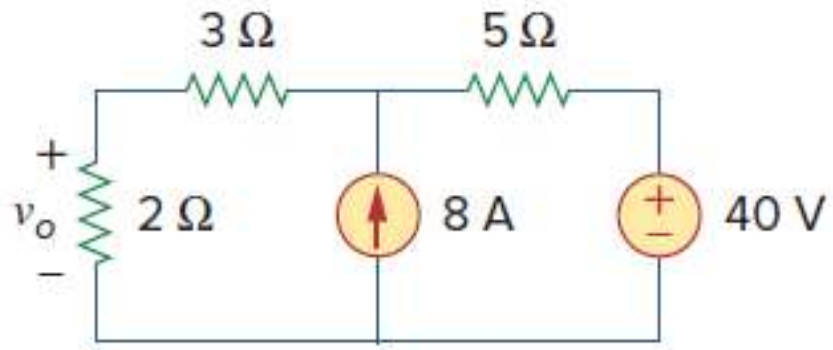


Ex 1: Use the superposition theorem to find “v” in the circuit



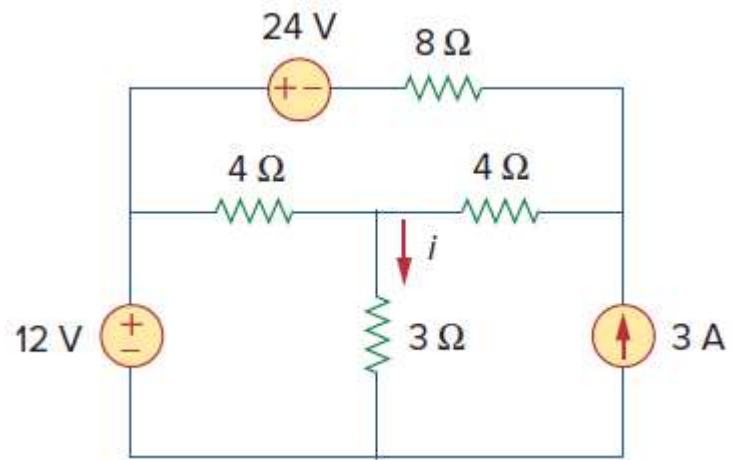
$$v = v_1 + v_2 = 2 + 8 = 10 \text{ V}$$

Problem 2 : Using the superposition theorem, find “ v_o ” in the circuit



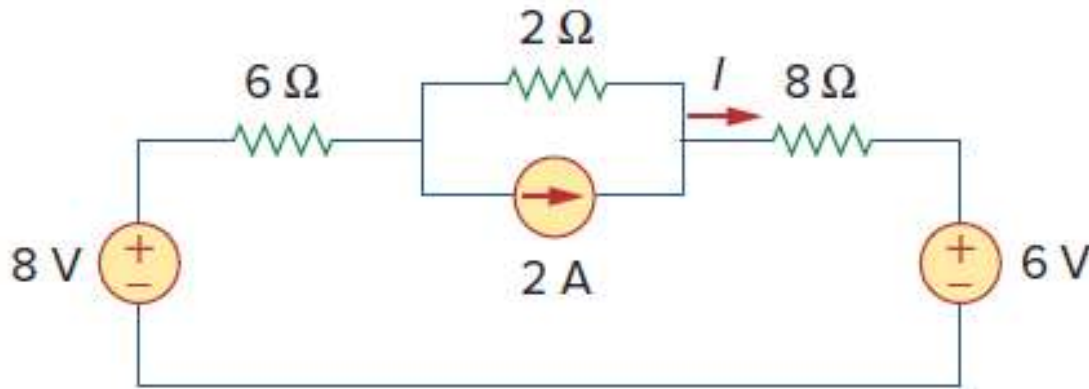
Answer: 16 V.

Problem 3 : Using the superposition theorem, find “i” in the circuit



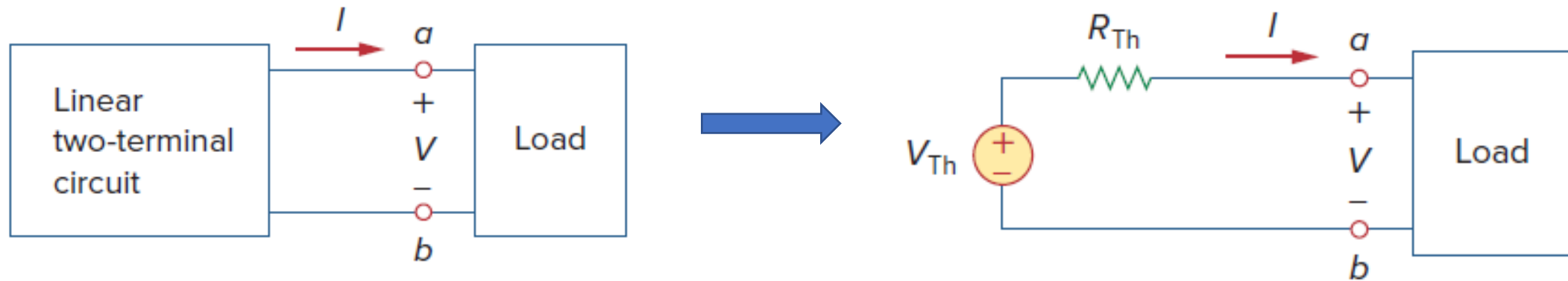
$$i = i_1 + i_2 + i_3 = 2 - 1 + 1 = 2 \text{ A}$$

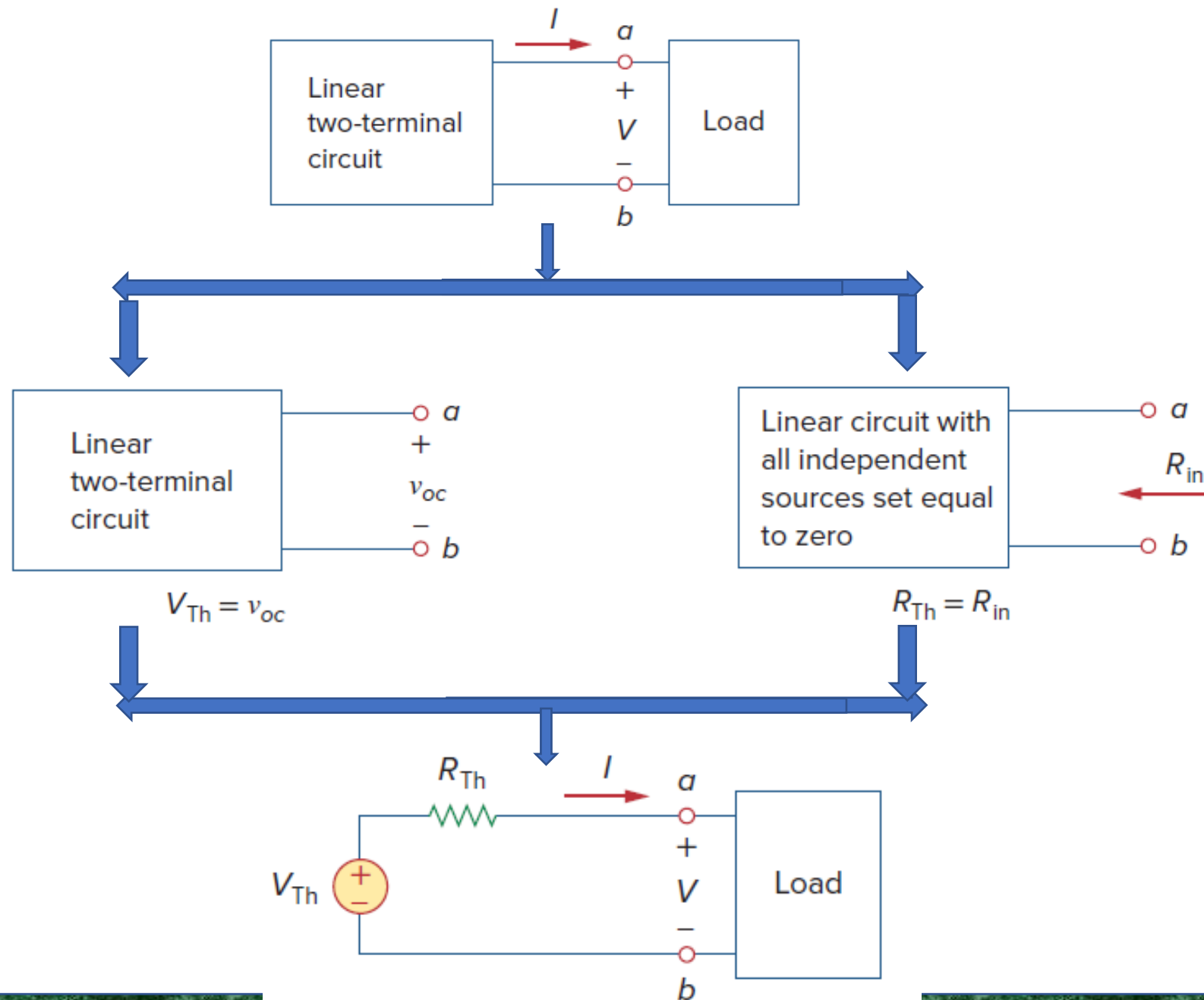
Problem 4: Using the superposition theorem, find “i” in the circuit

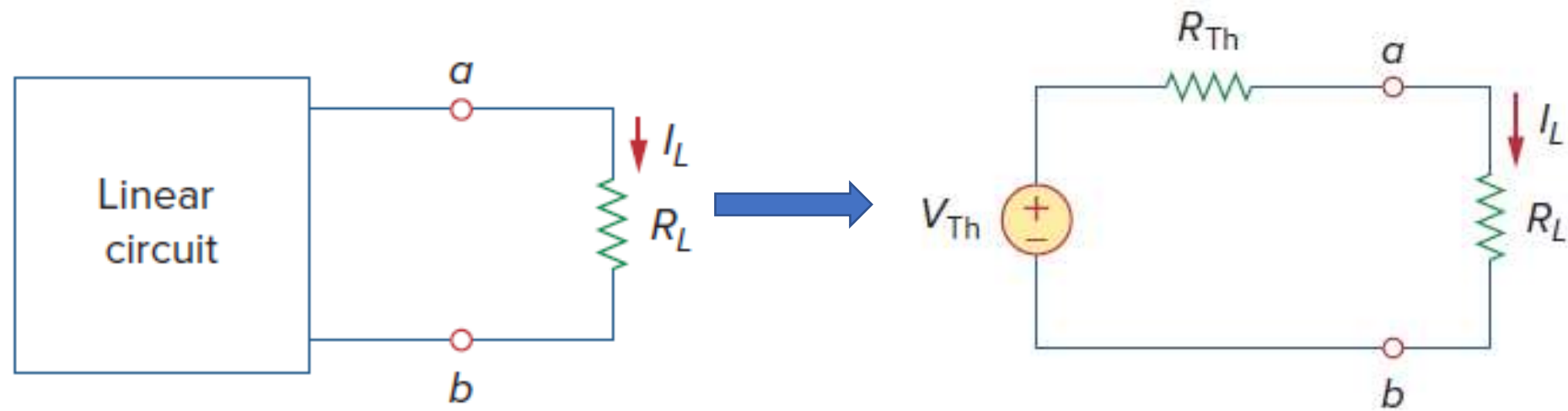


2.Thevenin's Theorem

- Thevenin's theorem states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a voltage source V_{Th} in series with a resistor R_{Th} .
- V_{Th} is the open-circuit voltage at the terminals and R_{Th} is the input or equivalent resistance at the terminals when the independent sources are turned off.

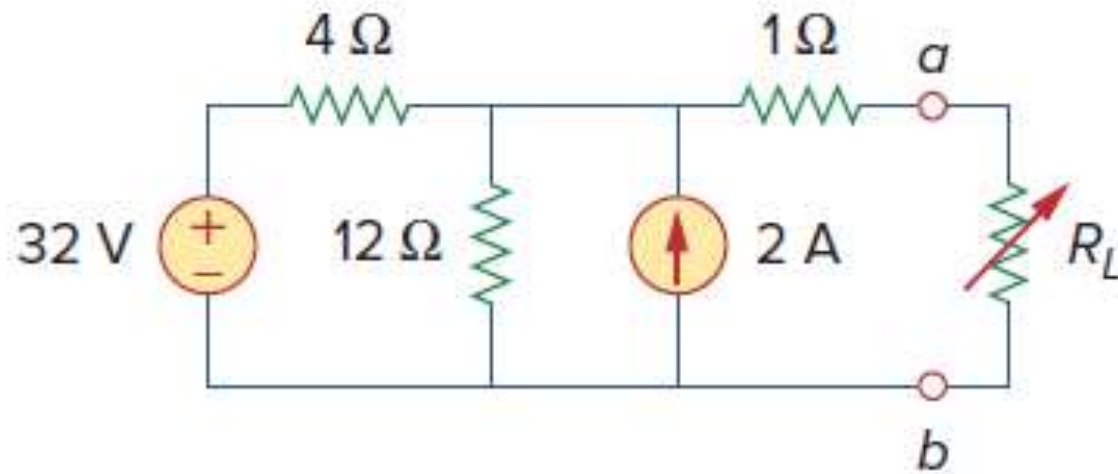




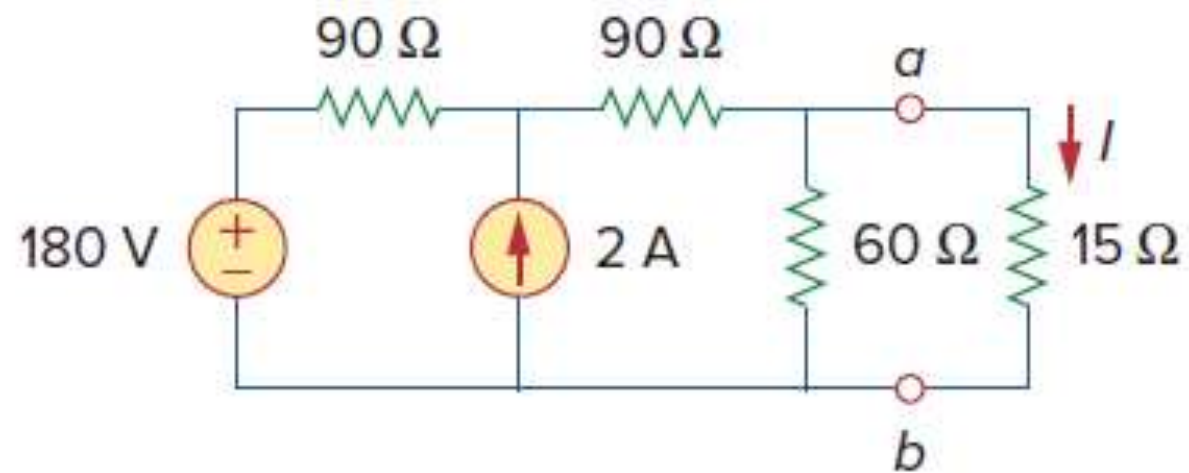


P-1

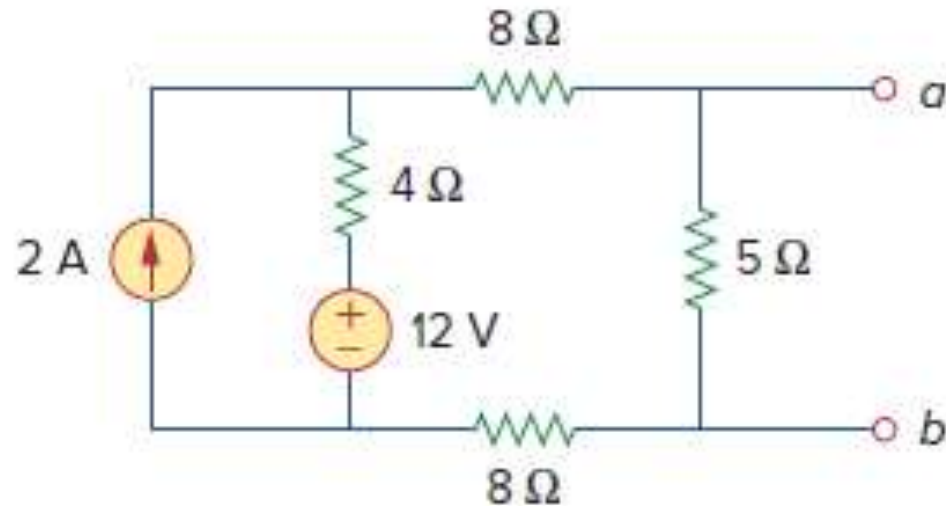
Find the Thevenin equivalent circuit of the circuit shown in Fig, to the left of the terminals a - b . Then find the current through $R_L = 10\ \Omega$, $15\ \Omega$, and $40\ \Omega$.



P-2: Find the Thevenin equivalent circuit of the circuit shown in Fig, to the left of the terminals a - b . Then find the current through $15\ \Omega$.



P-3: Find the Thevenin equivalent circuit of the circuit shown in Fig, to the left of the terminals a - b . Then find the current through if $R_L=10\ \Omega$ between a - b .



P-4: Find the Thevenin equivalent circuit of the circuit shown in Fig, to the left of the terminals a - b . Then find the current through $R_L=10\ \Omega$.

